When My Model Became the Question

How I Got to Where I'm Going

Tyler Hatstadt

(January 2025 - Present)

How I Got Here

I am drawn to systems that resist clean explanations. When a process is both highly studied and still difficult to explain, I instinctively want to understand why. This has led me toward problems where even defining the right question is part of the challenge.

Math and physics have taught me to look for underlying structure in complex systems, but the messy immune system fascinates me because its interactions are so difficult to disentangle, let alone explain. This curiosity led me to design an independent study in statistical immunology, where I encountered a puzzle I couldn't leave alone. Global Burden of Disease estimates suggested that rheumatoid arthritis incidence among adolescents and young adults had risen since 1990, even though genetic susceptibility was unlikely to have changed substantially over that period and smoking, a well-established risk factor, had declined in prevalence over the longer term. Meanwhile, obesity rates had risen sharply, particularly among younger people. I began wondering whether increasing adiposity could help explain some of the rise in young-onset disease. But how could I investigate that possibility when the relevant exposures might have occurred years or even decades before diagnosis, and the individual histories needed to examine them were largely unavailable?

When the independent study ended, I continued pursuing the question on my own. I initially tried regression models to identify patterns in the available data, but quickly ran into limitations that went beyond the scarcity of data. Young-onset RA datasets were limited, and the exposure histories that might matter most could precede diagnosis by years or decades. More fundamentally, observational data could not directly reveal the counterfactual outcomes I wanted to investigate. Around this time, I met Michael Bryan, an MD/PhD student at Oxford, whose approach to research resonated with me enough that I waited several months for the opportunity to work with him. He became my research mentor, and our discussions helped me think more critically about the limitations of the available data. I began to realize that waiting for the "perfect dataset" would not resolve the underlying inferential problem, so I decided to approach it through stochastic simulation. I knew simulation could neither reconstruct unobserved exposure histories nor establish causality, but it would allow me to make my assumptions explicit and explore how different exposure trajectories might affect population-level outcomes under those assumptions.

Eighteen months in, my research has progressed through two stages and is now entering a third. Stage I took me from a biological question to building a stochastic simulation to investigate it. Stage II began when I realized that the model itself had become something I needed to understand. I found myself asking not just what it predicted, but what its assumptions allowed me to infer and how much confidence I could place in its conclusions. Now, with recently secured access to longitudinal patient data, I am preparing to confront the model with real observations. I don't yet know where Stage III will lead. What interests me most is understanding why the model and the data might disagree, what those discrepancies could reveal about its assumptions, and what mathematical or statistical questions might emerge along the way.

When Building the Model Wasn't Enough

Once I committed to simulation, I began building the framework from the literature up. I built the simulation engine in Java as a Monte Carlo framework that generates synthetic populations of one million individuals, large enough to keep sampling noise small relative to the effects I wanted to examine. Rather than assigning fixed outcomes, I modeled individual risk probabilistically using published incidence estimates, odds ratios, and demographic factors, enabling me to compute prevalence patterns across age groups. I implemented an age-dependent exponential decay function to model attenuation of BMI-related risk over time and systematically perturbed obesity distributions to examine counterfactual population scenarios.

As an external consistency check, I compared predicted RA prevalence against Global Burden of Disease benchmarks and later used age-specific targets in calibration of selected model parameters. I also performed sensitivity analyses on key assumptions, including the BMI decay constant, to test whether plausible changes materially altered the conclusions. Each scenario was evaluated across 2,500 stochastic realizations, while uncertain parameters and assumptions were varied separately rather than treating repeated simulations as estimates of parameter uncertainty. As I refined the model, the differences between stochastic variability, parameter uncertainty, and model discrepancy became much clearer. Increasing the simulated population size or number of Monte Carlo realizations can reduce numerical uncertainty in estimated model outputs, but it cannot make uncertain parameters known or correct an inadequate model structure.

That understanding did not come cleanly. Several early versions of the model produced implausible outputs and unstable results. As the implementation grew more complicated, the code also became increasingly difficult to debug. Because this work was not tied to a grade or deadline, I was able to experiment freely, and that freedom proved essential. Rather than continuing to patch individual errors, I rebuilt the simulation several times from the ground up, making the framework increasingly modular and forcing myself to reconsider the assumptions embedded within it. In retrospect, many of the most important advances in my work came from the model's failures because they exposed gaps in my own reasoning. As the model improved, my discussions with Michael helped me gradually constrain a plausible range for BMI's modeled

population-level contribution to RA prevalence. Once the outputs stabilized within epidemiologically plausible bounds, the research began to feel less like constructing a model and more like understanding what the model could actually support.

That shift forced me to confront a broader methodological issue. Long-latency exposures are especially difficult because the relevant histories may be incomplete or entirely unobserved. Simulation cannot recover that missing information, and its conclusions remain conditional on the risk relationships, exposure distributions, and structural assumptions supplied to the model. What the model can do is make those assumptions explicit and test their consequences through counterfactual scenarios. I began to recognize that reproducing plausible population-level patterns was not the same as establishing that the assumptions generating those patterns were correct.

After several months of revising the model, I posted a manuscript to bioRxiv and submitted an abstract to the American Association of Immunologists (AAI) Conference. After scientific review, I was accepted to present and was the conference's only high school student presenter. Preparing the work for a professional audience forced me to compress months of modeling decisions into a clear argument, which sharpened both my conclusions and my understanding of the framework's assumptions and limitations. At the conference, I presented the model to clinical researchers and fielded questions about its structure, interpretation, and potential beyond rheumatoid arthritis. Those discussions pushed me to think more carefully about which aspects of the work were specific to the biological application and which might reflect a broader methodological problem.

When Validation Became an Inference Problem

Although the conference expanded my thinking, my efforts to move beyond simulation had begun much earlier. For many months, I had been reaching out to clinical researchers for longitudinal patient data with the hope of validating the simulator by comparing its predictions against observed patient outcomes. After multiple proposals, rejections, and calls, I received approval to work with de-identified patient data from Seattle Children's Hospital as well as the ERAS–ERAN cohorts through King's College London. The well-established, multicenter ERAS–ERAN cohorts stood out because of the scale, depth, and duration of their longitudinal records, encompassing more than 2,700 patients with rheumatoid arthritis and follow-up extending to 25 years. My independent proposal was approved by the cohorts' Steering Group around the time of the conference, bringing the prospect of empirical validation into focus just as my conversations with other researchers were expanding the questions I wanted to investigate.

My conversations with scientists at the conference broadened my understanding of the problem I was studying and left me with a growing sense that the questions I had been asking were only a small part of a much larger intellectual landscape. I found myself increasingly curious about what remained unanswered and began to wonder whether my original validation objective could actually establish that the model was correct. I realized that agreement with real observations would not necessarily establish that its underlying assumptions were valid. Just as importantly, disagreement with the data would not by itself reveal *why* the model might fail. It could reflect poorly estimated parameters, stochastic variation, measurement error, omitted structure, or broader model misspecification. My thinking gradually transitioned to a more difficult question about what could actually be inferred from incomplete observations.

I began treating the framework less as a biological simulator and more as a mathematical object in its own right. This meant examining which quantities were observable, which underlying states and exposure histories remained latent, and how assumptions about risk relationships influenced what could be inferred. I became particularly interested in whether the underlying states and parameters could be recovered from partial longitudinal observations, whether that recovery would be identifiable, and how sensitive it might be to incomplete or noisy observations. If different latent histories, parameter combinations, or model structures could produce essentially the same observations, additional computation alone could not determine which had generated the real system.

I began rebuilding the computational framework around these new inference questions. Rather than continue modifying the original simulator, I preserved that codebase as a record of the completed research and created a second repository, *longitudinal-risk-inference*, to develop the new framework from scratch. The original model was designed primarily to estimate population-level counterfactual differences, whereas the new simulator follows individuals through annual transitions involving disease incidence, mortality, evolving BMI trajectories, and changing risk states. Instead of relying primarily on static population summaries, the new framework generates individual trajectories whose observable components can eventually be compared with available longitudinal measurements. Because the data-generating process is known, the simulator provides a controlled environment for testing inference procedures before applying them to real observations. Rebuilding the model this way made the distinction between latent states and observable outcomes more explicit. For example, changes in the assumed BMI-risk relationship and changes in underlying BMI trajectories could produce similar population-level incidence patterns. Longitudinal observations may help distinguish these explanations, but whether they contain sufficient information to do so remains an open question.

One preliminary experiment asks whether a flexible machine-learning model can identify patterns consistent with relationships deliberately encoded in the simulator without being given their functional form. I trained a gradient-boosted model on simulated individual-level outcomes and used feature and interaction analyses to investigate whether it could identify the encoded risk relationships, including the age-dependent BMI effect. Because the training data are generated by my own simulator, identifying these patterns is not biological evidence. Rather, the experiment provides a controlled test of whether the modeling and analysis pipeline can detect known patterns in simulated data.

The more difficult question is what happens when similar methods are applied to real longitudinal data, where the generating mechanism is unknown, observations may be incomplete, and the model itself may be wrong.

This work has also led me to reconsider an assumption that was necessary in the original simulator. Without sufficient individual-level observations, I had to specify many of the model's risk relationships using published epidemiological evidence. Access to longitudinal data creates the possibility of estimating some of those relationships directly rather than assuming them in advance. I am increasingly interested in understanding which components of a stochastic model should remain mechanistically specified, which might be learned from observations, and how those choices affect the reliability and interpretability of the resulting inferences. What began as an effort to validate a biological simulation has led me toward a broader question about how much can actually be learned about a complex stochastic system when its underlying mechanisms are only partially observed.

Where the Questions Lead Next

The next stage of my research is to use real longitudinal observations to find where the framework's assumptions or structure begin to break down. My first challenge will be determining which aspects of the simulator can meaningfully be evaluated against the available ERAS–ERAN observations. Where comparisons are possible, I want to investigate whether discrepancies reflect uncertain parameters, stochastic variation, weakly identifiable latent structure, model misspecification, or relationships that might be better estimated from data than imposed mechanistically.

I do not yet know which of these limitations will prove most important, and I do not want to choose a mathematical method before the data reveal what problem actually needs to be solved. That might mean confronting an identifiability problem, investigating systematic model discrepancy, or determining whether certain mechanistic relationships would be better learned from data. I therefore want to understand not only whether the model fits, but what aspects of the underlying system the data actually constrain and which conclusions remain dependent on assumptions that the observations cannot resolve. I want the mathematical direction of this stage to emerge from the limitations exposed by the data rather than forcing the data into a problem I have already decided to solve.

What matters to me is not making the framework mathematically more elaborate for its own sake, but identifying a limitation whose resolution requires a genuinely new method or result. Rheumatoid arthritis provides a concrete system in which those limitations can be exposed and tested, but I increasingly see it as a testbed for broader mathematical questions as well. If the same inferential difficulty arises from partial observation, latent structure, model misspecification, or the interaction between mechanistic and learned components, then the problem may extend well beyond this particular disease. At the same time, I want this work to bring me closer to the question that originally motivated my research, whether rising adiposity could plausibly contribute to young-onset RA.

After 18 months of sustained work, this process has fundamentally changed my understanding of both research and my own reasoning. I began with a question that felt logically inconsistent, assuming the primary challenge would be analytical. Instead, I encountered unstable models, flawed assumptions, data access barriers, and periods of wondering where to go next. I gradually learned that making progress required more than finding solutions to difficult problems. It often meant recognizing that the assumptions guiding my work were themselves inadequate, and being willing to reconsider questions I had spent months trying to answer. Much of this work was independent, which meant learning to recognize weaknesses in my own reasoning without the constant exchange of ideas I had experienced in other collaborative work. I came to understand that meaningful work is rarely a linear progression of correct ideas, but an iterative process in which errors expose hidden weaknesses and force clearer thinking.

My experience with independent research has also made me appreciate how much my approach to difficult problems was shaped by 2.5 years on my competitive physics team. During that time, I spent four afternoons each week working alongside other students and PhD instructors on highly complex, open-ended physics problems that often took an entire year to investigate. I remember the whiteboards, the experiments, and the constant movement of people in and out of conversations as we tried to make progress on questions none of us knew how to solve. Younger students were treated as genuine contributors, while our instructors helped us explore unfamiliar approaches without taking ownership of the problems away from us. We could hardly wait to return each day. That environment taught me to develop ideas through discussion, criticism, and shared experimentation. Working largely independently on the RA project challenged me in a different way, requiring me to cultivate that same willingness to question my reasoning even when there was no one beside me at the whiteboard.

As the questions have grown more difficult, I have found myself increasingly drawn to the uncertainty of open-ended research. There is something deeply satisfying about trying approaches that fail, reconsidering assumptions, and gradually finding a way forward when the answer is not apparent. I still cannot let go of my original nagging question about whether rising adiposity could plausibly contribute to young-onset RA, but pursuing it has opened questions I never expected to encounter and may continue exploring for years. What began as an effort to understand one apparent inconsistency has changed how I think about problems, how I respond when my reasoning falls short, and what I find rewarding about research itself. More than anything, though, my work is driven by a simple, stubborn desire to understand how things work.